SPC quick reference

IES | Primer 3.0 | Synthetic worked example

Statistical Process Control compares time-ordered observations with a baseline. A control chart describes behaviour; specifications state requirements.

Worked individuals chart

For 30 baseline readings: mean=10.01333 L; average moving range=0.18621 L. Sigma estimate=MR average/1.128=0.16508 L. Limits are mean +/- 3 sigma: LCL=9.51810 L, UCL=10.50856 L.

Signal

Monitoring sample 33 is 10.8 L and exceeds the baseline UCL. Investigate; do not redraw limits to conceal it or infer a physical cause.

Capability illustration

Using teaching specifications 9.4-10.6 L: Cp=(USL-LSL)/(6 sigma)=1.212. Cpk=min(USL-mean,mean-LSL)/(3 sigma)=1.185.

Conditions

Capability interpretation requires stable behaviour, suitable measurement and appropriate distribution and independence assumptions. This small training dataset does not certify capability. Source: NIST/SEMATECH e-Handbook; see REFERENCES.md.